

Array List

Vectors $\leftarrow$ C++

// Syntax

Initial Capacity

```
ArrayList < Datatype > list = new ArrayList < > ( );
```

↑
Class

Starts with capital letter

Array

ArrayList

```
int [] nums = new int [5];
```

```
ArrayList < Integer > nums = new ArrayList
```

// adding elements to AL

① `nums.add(5);` ✓

② `System.out.println(list.contains(654));`

③ // updating list

④ `list.set(index, value)` ✓

④ `list.remove(index)`

⑤ Iterating

```
for (i=0; i < n; i++)
```

```
list.add(list.get(i));
```

```
}
```

List can contain more elements than its size

④ Resizing Mechanism of ArrayList

- ① new array list is created with larger capacity
- ② Element from old is copied to new
- ③ Old $\rightarrow$ Garbage collection (remove)

new capacity = Old Capacity + 50% of Old Capacity

Capacity	New Cap
10	
10 + 5	15
15 + 7	21